

# PRINCE CHAUHAN

## SENIOR QUANTITATIVE ANALYST & RESEARCHER

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SEBI Registered Research Analyst

### PROFESSIONAL SUMMARY

Derivatives research professional with 9+ years of experience spanning options trading, market analysis, and systematic strategy development. Combines deep institutional derivatives expertise with AI-augmented quantitative engineering to design, prototype, and rigorously validate production alpha and execution pipelines across Indian equity derivatives and US index options. Experienced in strategy validation, portfolio risk monitoring, and automated research workflows.

### PROFESSIONAL EXPERIENCE

#### SMC Global Securities Ltd. - Senior Quantitative Analyst & Researcher

Dec 2024 - Present

- **Alpha Research Infrastructure:** Architected event-driven simulation pipelines and point-in-time research infrastructure on tick-level and 1-minute historical bar feeds across US and Indian index derivatives (SPX, VIX, Nifty, Bank Nifty).
- **US Index Derivatives:** Engineered automated execution pipelines and volatility surface modeling for short-dated US index options (SPX/SPXW) via Interactive Brokers (IBKR API), systematically capturing short-horizon IV skew mispricings with dynamic delta rebalancing.
- **Statistical Arbitrage:** Formulated Bank Nifty constituent pair and basket models applying Johansen cointegration rank tests, dynamic rolling beta hedging, and spread mean-reversion half-life estimation to systematically capture relative-value opportunities.
- **Chronological Walk-Forward Backtesting:** Evaluated systematic volatility strategies across Nifty and Bank Nifty index options over a 7-year sample (2018-2024); achieved an annualized out-of-sample Sharpe of 2.5+ and profit factor of 2.1+ after deducting 0.50% round-trip option premium turnover friction (covering STT, exchange fees, and bid-ask crossing slippage).
- **Execution Architecture & Telemetry:** Built pre/post-trade Transaction Cost Analysis (TCA) and live execution telemetry, reducing market crossing drag by 24% and eliminating broker order-state desynchronization and duplicate executions via atomic client order IDs.
- **Desk Mentorship & Risk Alignment:** Mentored 3 junior quantitative researchers and traders in statistical validation methods and execution best practices; partnered with Portfolio Managers on strategy scalability, real-time portfolio Greeks, dealer GEX mapping, parametric VaR, and SPAN margin caps.

#### Smart Bulls Capital - Research Analyst, Derivatives & Quantitative Research

Mar 2017 - Nov 2024

- **Options Volatility Books:** Researched, structured, and traded systematic multi-leg options volatility strategies (Iron Condors, Butterflies, Calendar Spreads) with dynamic delta/vega rebalancing guided by real-time IV surface spline smoothing.
- **Mathematical Root-Finding Solvers:** Computed implied volatility by numerically inverting Black-Scholes option prices using Newton-Raphson and Brent root-finding algorithms; mapped dealer Gamma Exposure (GEX) profiles to quantify institutional pin risk.
- **Portfolio Construction & Allocation:** Implemented Risk Parity and Mean-Variance capital allocation models across multi-strike options books; managed daily P&L attribution, SPAN margin utilization, and drawdown limits within desk risk budgets.
- **Fundamental Balance-Sheet Gating:** Conducted forensic financial statement audits, DCF valuation, and working-capital variance analysis across banking and industrial equities to gate capital deployment to quantitative models.

#### Earlier Corporate Experience - Corporate Finance & Audit Roles

2014 - 2017

- **Corporate Treasury & Audit:** Executed treasury operations and working-capital forecasting at Indian Oil Corporation Ltd. (IOCL, 2016-2017); conducted statutory audits and corporate tax structuring at CA Matta & Associates (2014-2016).

## SELECTED INDEPENDENT QUANTITATIVE RESEARCH (OPEN SOURCE)

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Hosted at: [github.com/prince-chauhan-pc/quant-research-portfolio](https://github.com/prince-chauhan-pc/quant-research-portfolio)

### Point-in-Time Backtesting Framework & Invariant Gate (Python)

2024

- **Research Objective:** Enforce point-in-time data alignment and realistic transaction friction in high-turnover options strategy research.
- **Methodology & Validation:** Implemented rolling walk-forward optimization partitioned into 70% in-sample parameter calibration, 20% out-of-sample cross-validation, and 10% held-out test windows to prevent data leakage, paired with 1,000-path Monte Carlo drawdown stress testing.
- **Audit Guardrails:** Engineered an automated data-integrity test suite enforcing Rules L1-L17 across 22 boundary checks, pre-deducting 0.50% round-trip premium friction and exchange SPAN margins.

### Arbitrage-Free SVI Volatility Surface & GEX Profiling

2024

- **Research Objective:** Eliminate calendar and butterfly arbitrage from market implied volatility quotes for robust cross-strike option pricing.
- **Methodology:** Calibrated daily implied volatility surfaces using SVI parameterization with non-linear optimization; enforced butterfly convexity and calendar-spread non-arbitrage constraints.
- **Practical Outcome:** Stabilized cross-strike smile interpolation across illiquid strikes and generated real-time dealer GEX sensitivity curves to map institutional pin-risk zones.

### Bank Nifty Basket Cointegration & Statistical Arbitrage

2024

- **Research Objective:** Systematically exploit stationary mean-reverting spreads among top banking constituent equities.
- **Methodology & Risk:** Applied Johansen multivariate cointegration tests to identify stationary linear combinations; estimated spread mean-reversion half-life to calibrate entry/exit thresholds and holding periods, maintaining delta neutrality via rolling OLS beta weights.

### Microstructure Execution Simulator & TCA Engine

2024

- **Research Objective:** Accurately quantify market crossing costs and adverse selection for algorithmic limit orders.
- **Methodology:** Modeled Limit Order Book (LOB) queue priority, non-linear market impact, and fill probability across market regimes classified by Parkinson realized volatility.

## TECHNICAL & QUANTITATIVE COMPETENCIES

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- **Quantitative Research:** Statistical Arbitrage, Johansen Cointegration, Dynamic Beta Hedging, SVI Volatility Surface Fitting, Parkinson Realized Volatility, Market Regime Classification (K-Means), Walk-Forward Optimization, Monte Carlo Simulation.
- **Derivatives & Risk:** Black-Scholes Pricing, Portfolio Greeks (Delta, Gamma, Vega, Theta), Dealer GEX Profiling, Parametric & Historical VaR, SPAN Cross-Margin Optimization, Risk Parity, Mean-Variance Allocation.
- **AI-Augmented Engineering:** Agentic LLM Architecture for Rapid Alpha Prototyping, Automated Pipeline Construction, Synthetic Scenario Generation, Code Verification, and Mathematical Invariant Auditing.
- **Core Languages & Systems:** Python (NumPy, SciPy, Polars), SQL (DuckDB, PostgreSQL), Apache Parquet, Git, Linux.
- **Markets & Microstructure:** Indian Equity Derivatives (NSE Nifty, Bank Nifty, Sensex), US Equity Derivatives (CBOE SPX, VIX), Pre/Post-Trade TCA, Limit Order Book Queue Dynamics, Smart Order Slicing.

## EDUCATION & REGULATORY CREDENTIALS

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- **Bachelor of Commerce (Honours):** Kirori Mal College, University of Delhi (2011 - 2014) | First Division (71% Marks)
- **Chartered Accountancy (CA Final):** Institute of Chartered Accountants of India (ICAI, 2018) | Cleared Group 1 (78% Marks in Financial Reporting & Auditing)
- **Regulatory Licenses:** SEBI Registered Research Analyst | NISM Series-XV